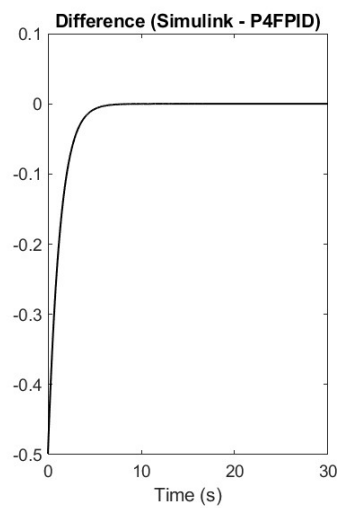
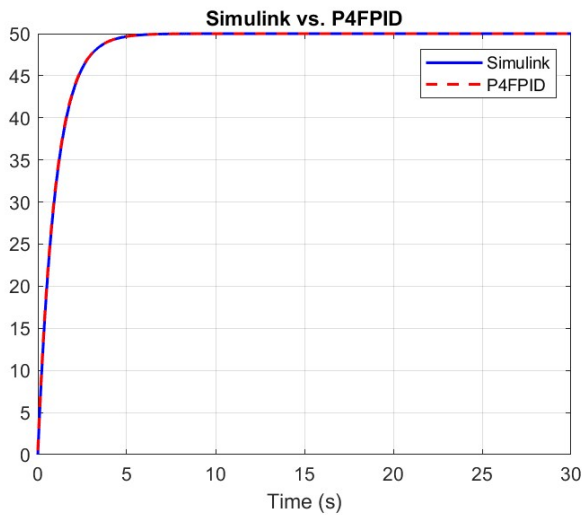


P4FPID Software Evaluation

This document presents a series of comparative simulation experiments carried out to evaluate the consistency and validity of the results produced by the developed software P4FPID with respect to those obtained from MATLAB Simulink under the default setting of the fuzzy controller which is 1000 discretization point. The study explores several configurations, including open loop, closed loop, and fuzzy control schemes, under different controller structures and simulation step sizes. The experiments were conducted using the Runge–Kutta fourth order (RK4) numerical integration method, which was selected by the authors as it represents one of the most commonly adopted methods in control simulation environments. It is important to note that this report is not intended for peer review publication, but rather serves as a technical validation document that accompanies the software distribution. All relevant supplementary materials including MATLAB scripts, Simulink simulation models, and Fuzzy Logic Toolbox configuration files are provided to ensure transparency and reproducibility of the presented results. Through these materials and visual comparisons of the generated response graphs, readers can assess the capability of P4FPID to produce reliable and coherent simulation outcomes under varying control and system conditions. It is also worth noting that, in the discrete-time domain, the outputs of both systems are found to be perfectly identical, and therefore no further comparison between the two environments was deemed necessary in this case.

Opened-Loop Simulation

Simulation Step 0.01



Results:

Fixed Step Time = 0.01 sec

Algorithm: RK-4 th order

RMSE = 0.0731

MRE = 0.1860%

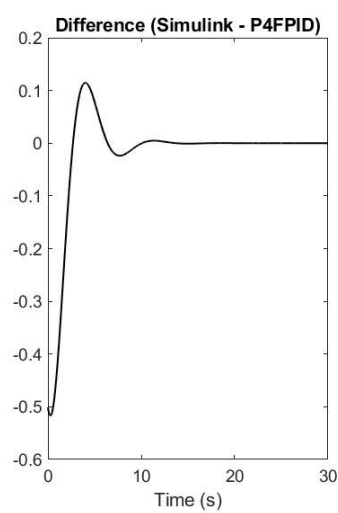
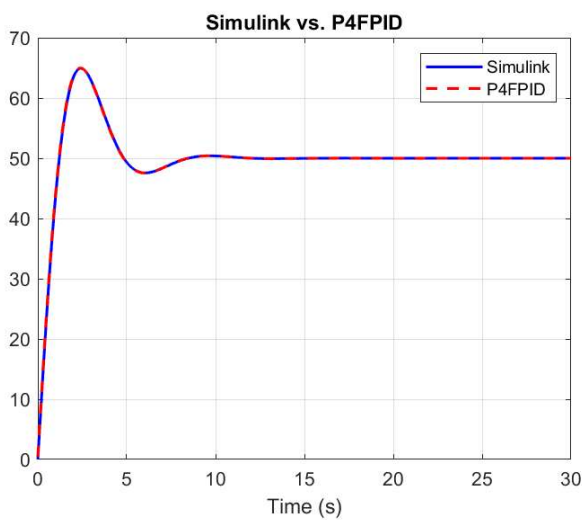
Correlation = 1.0000

Transfer Function

$$G(s) = \frac{1}{1s + 1}$$

Put in: Opened-Loop

CSV File: [1.0];[1.0;1.0].csv



Results:

Fixed Step Time = 0.01 sec

Algorithm: RK-4 th order

RMSE = 0.1102

MRE = 0.2175%

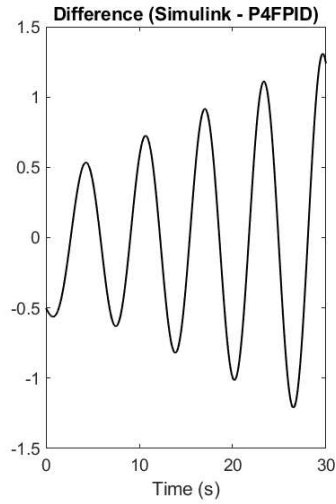
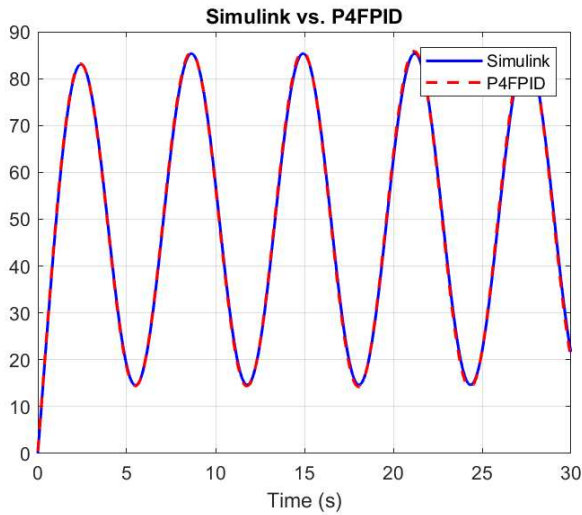
Correlation = 0.9999

Transfer Function

$$G(s) = \frac{1s + 1}{1s^2 + 1s + 1}$$

Put in: Opened-Loop

CSV File: [1.0;1.0];[1.0;1.0;1.0].csv



Results:

Fixed Step Time = 0.01 sec

Algorithm: RK-4 th order

RMSE = 0.6497

MRE = 1.5791%

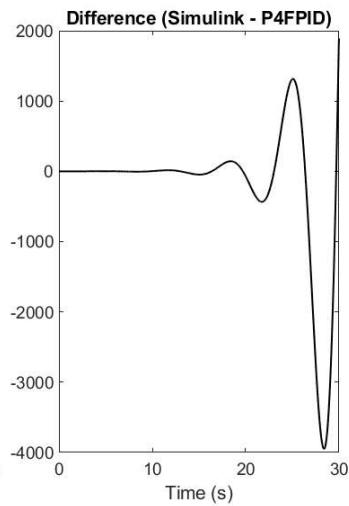
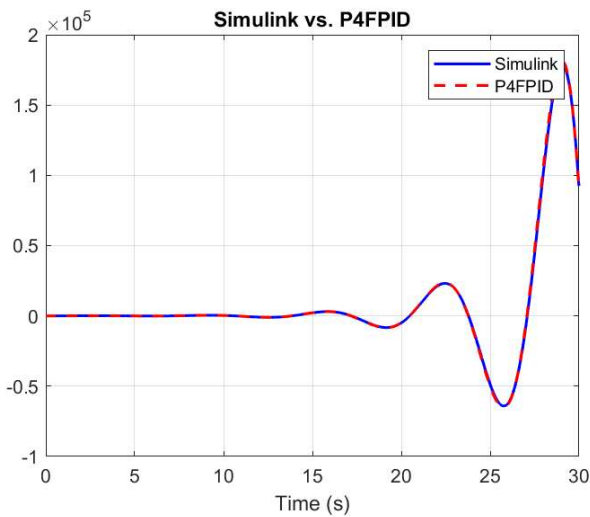
Correlation = 0.9997

Transfer Function

$$G(s) = \frac{1s^2 + 1s + 1}{1s^3 + 1s^2 + 1s + 1}$$

Put in: Opened-Loop

CSV File: [1.0;1.0;1.0];[1.0;1.0;1.0;1.0].csv



Results:

Fixed Step Time = 0.01 sec

Algorithm: RK-4 th order

RMSE = 972.0803

MRE = 6.4540%

Correlation = 0.9999

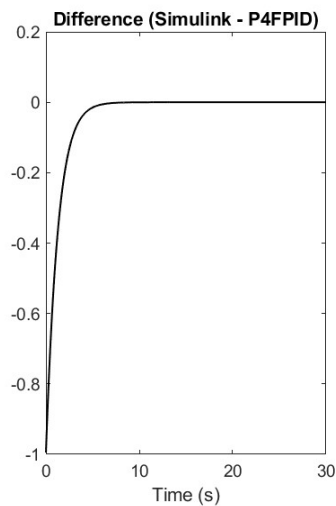
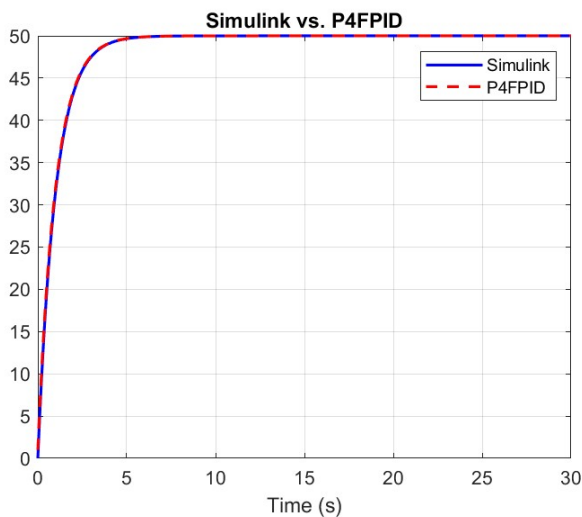
Transfer Function

$$G(s) = \frac{1s^3 + 1s^2 + 1s + 1}{1s^4 + 1s^3 + 1s^2 + 1s + 1}$$

Put in: Opened-Loop

CSV File: [1.0;1.0;1.0;1.0];[1.0;1.0;1.0;1.0;1.0].csv

Simulation Step 0.02



Results:

Fixed Step Time = 0.02 sec

Algorithm: RK-4 th order

RMSE = 0.1463

MRE = 0.3251%

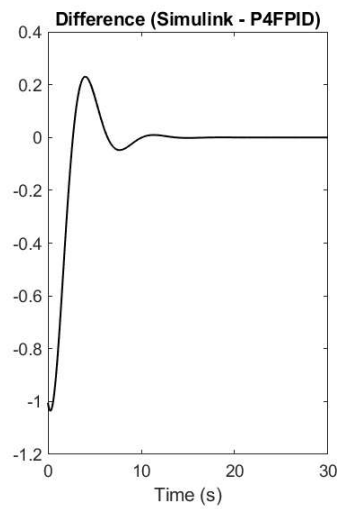
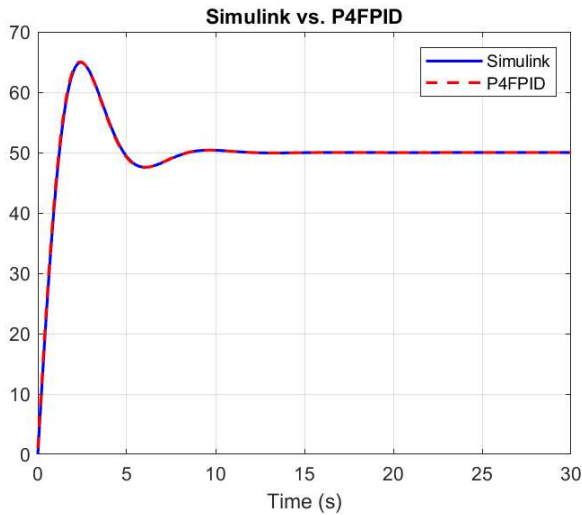
Correlation = 1.0000

Transfer Function

$$G(s) = \frac{1}{1s + 1}$$

Put in: Opened-Loop

CSV File: [1.0];[1.0;1.0].csv



Results:

Fixed Step Time = 0.02 sec

Algorithm: RK-4 th order

RMSE = 0.2205

MRE = 0.3890%

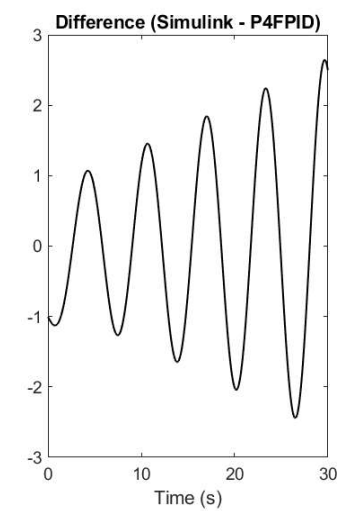
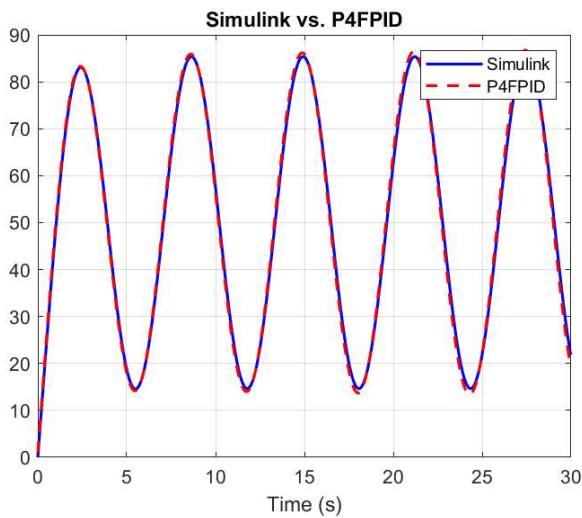
Correlation = 0.9996

Transfer Function

$$G(s) = \frac{1s + 1}{1s^2 + 1s + 1}$$

Put in: Opened-Loop

CSV File: [1.0;1.0];[1.0;1.0;1.0].csv



Results:

Fixed Step Time = 0.02 sec

Algorithm: RK-4 th order

RMSE = 1.3094

MRE = 3.1269%

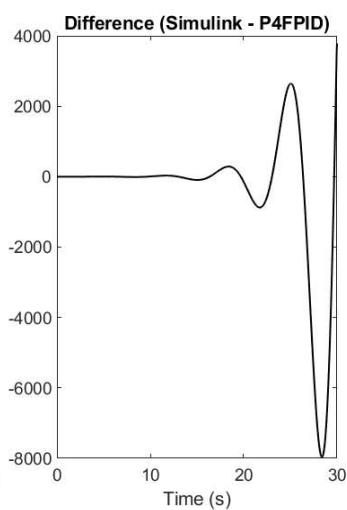
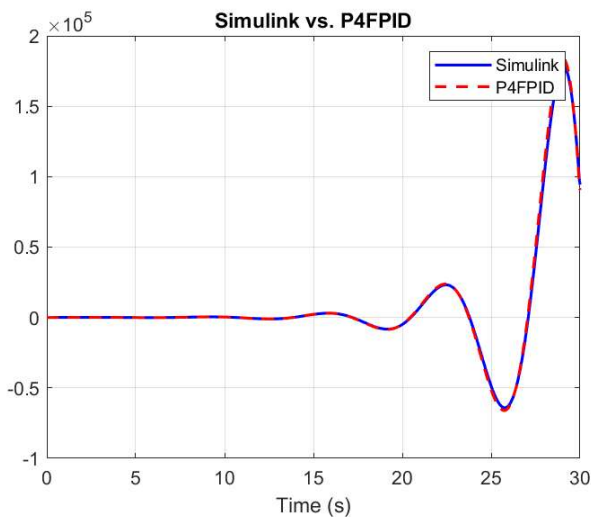
Correlation = 0.9989

Transfer Function

$$G(s) = \frac{1s^2 + 1s + 1}{1s^3 + 1s^2 + 1s + 1}$$

Put in: Opened-Loop

CSV File: [1.0;1.0;1.0];[1.0;1.0;1.0;1.0].csv



Results:

Fixed Step Time = 0.02 sec

Algorithm: RK-4 th order

RMSE = 1960.6338

MRE = 12.1605%

Correlation = 0.9996

Transfer Function

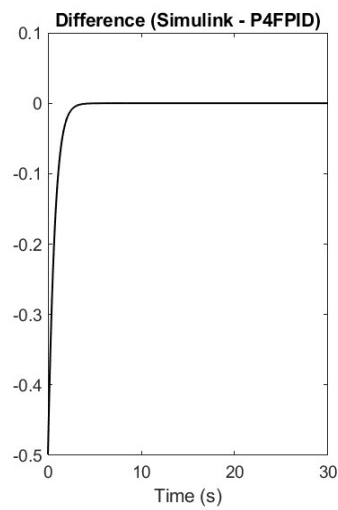
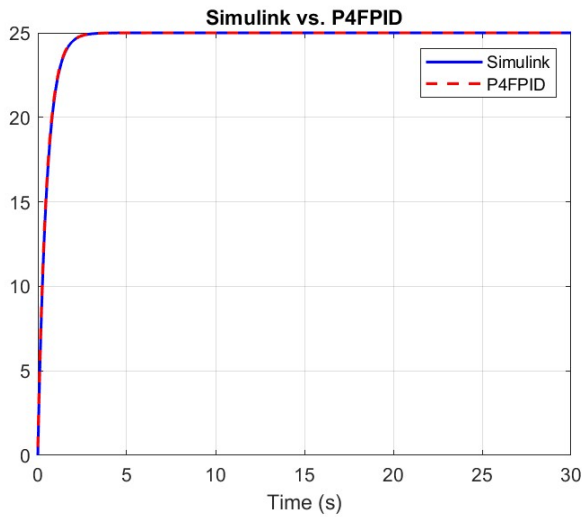
$$G(s) = \frac{1s^3 + 1s^2 + 1s + 1}{1s^4 + 1s^3 + 1s^2 + 1s + 1}$$

Put in: Opened-Loop

CSV File: [1.0;1.0;1.0;1.0];[1.0;1.0;1.0;1.0;1.0].csv

Closed-Loop Simulation

Simulation Step 0.01



Results:

Fixed Step Time = 0.01 sec

Algorithm: RK-4 th order

RMSE = 0.0550

MRE = 0.1696%

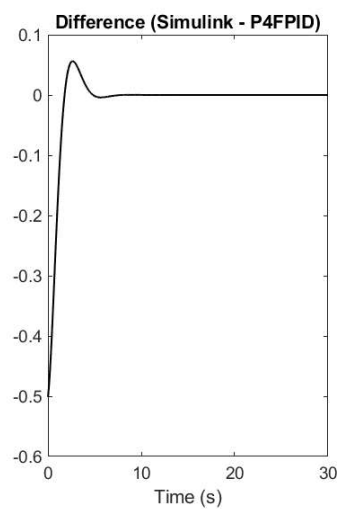
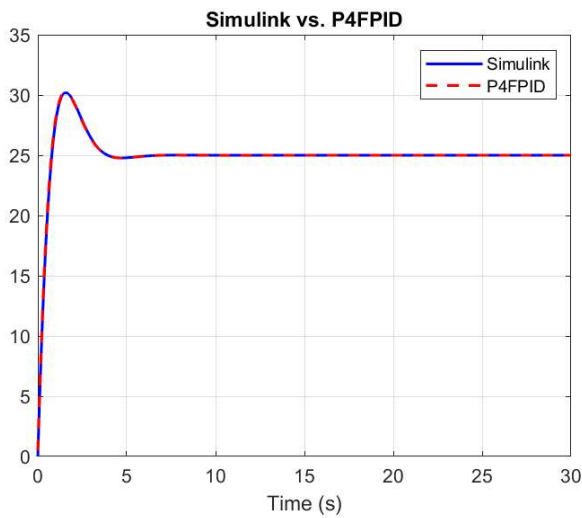
Correlation = 1.0000

Transfer Function

$$G(s) = \frac{1}{1s + 1}$$

Put in: Closed-Loop

CSV File: [1.0];[1.0;1.0].csv



Results:

Fixed Step Time = 0.01 sec

Algorithm: RK-4 th order

RMSE = 0.0715

MRE = 0.1917%

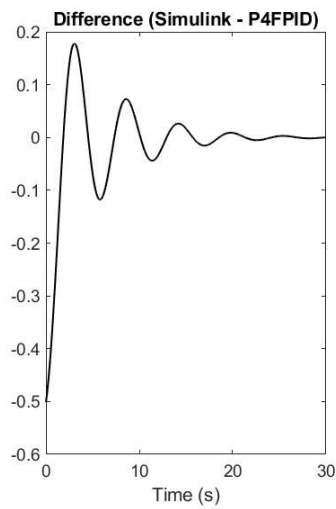
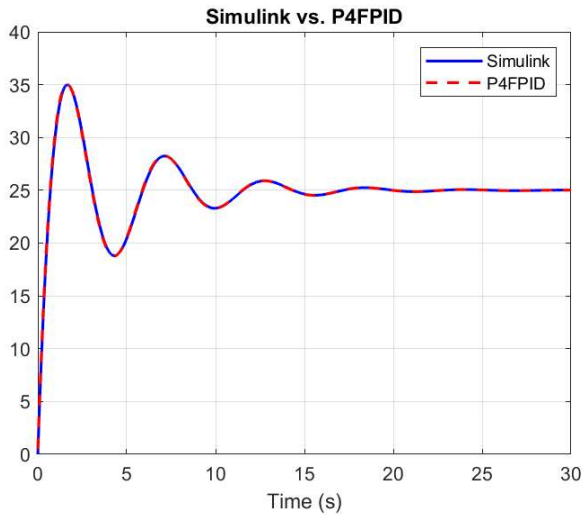
Correlation = 0.9998

Transfer Function

$$G(s) = \frac{1s + 1}{1s^2 + 1s + 1}$$

Put in: Closed-Loop

CSV File: [1.0;1.0];[1.0;1.0;1.0].csv



Results:

Fixed Step Time = 0.01 sec

Algorithm: RK-4 th order

RMSE = 0.0931

MRE = 0.2990%

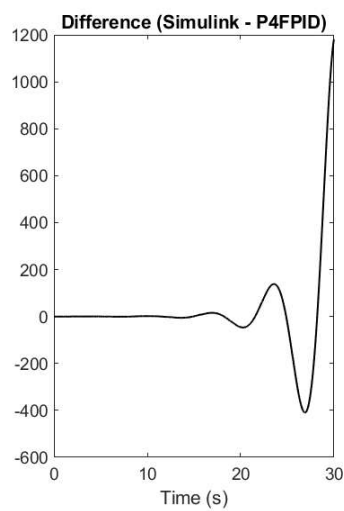
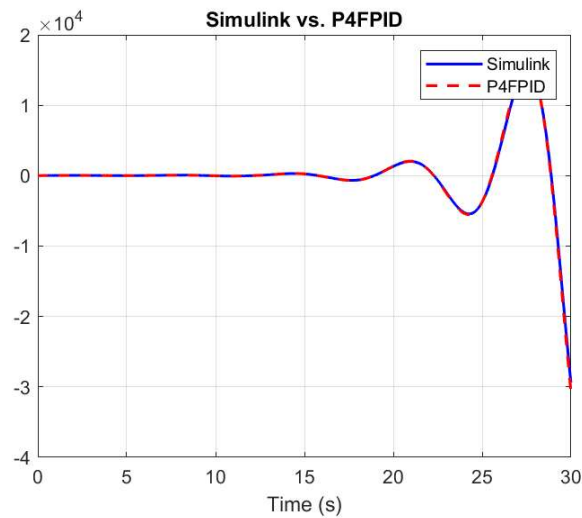
Correlation = 0.9996

Transfer Function

$$G(s) = \frac{1s^2 + 1s + 1}{1s^3 + 1s^2 + 1s + 1}$$

Put in: Closed-Loop

CSV File: [1.0;1.0;1.0];[1.0;1.0;1.0;1.0].csv



Results:

Fixed Step Time = 0.01 sec

Algorithm: RK-4 th order

RMSE = 207.0445

MRE = 11.8879%

Correlation = 0.9997

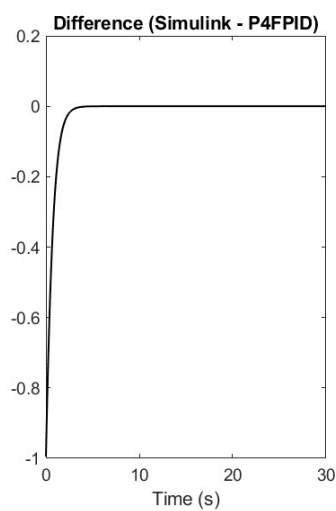
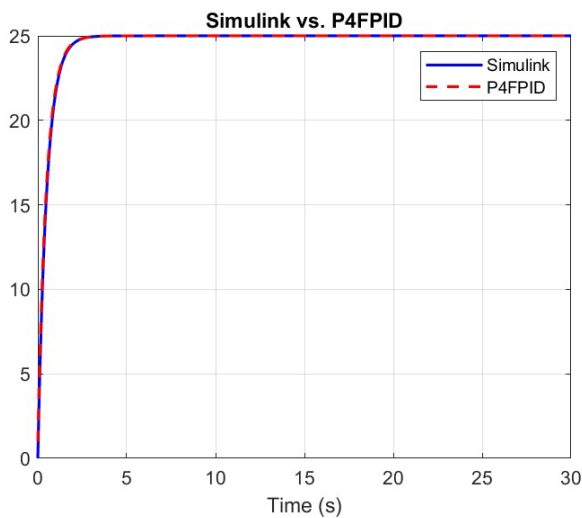
Transfer Function

$$G(s) = \frac{1s^3 + 1s^2 + 1s + 1}{1s^4 + 1s^3 + 1s^2 + 1s + 1}$$

Put in: Closed-Loop

CSV File: [1.0;1.0;1.0;1.0];[1.0;1.0;1.0;1.0;1.0].csv

Simulation Step 0.02



Results:

Fixed Step Time = 0.02 sec

Algorithm: RK-4 th order

RMSE = 0.1103

MRE = 0.2921%

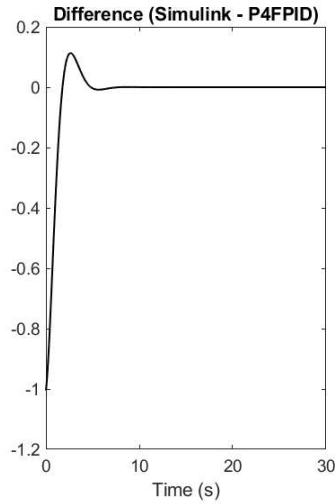
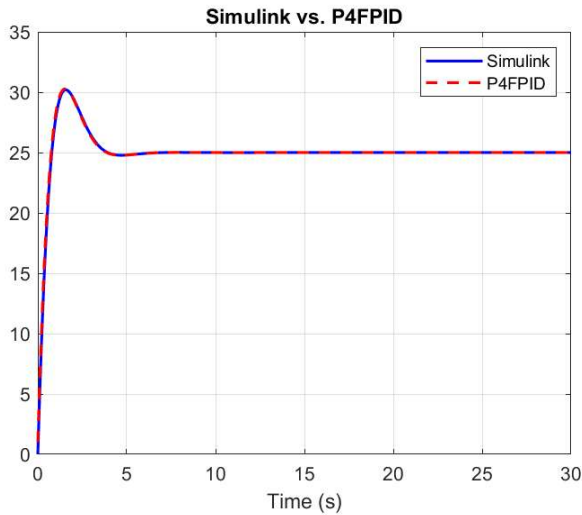
Correlation = 1.0000

Transfer Function

$$G(s) = \frac{1}{1s + 1}$$

Put in: Closed-Loop

CSV File: [1.0];[1.0;1.0].csv



Results:

Fixed Step Time = 0.02 sec

Algorithm: RK-4 th order

RMSE = 0.1435

MRE = 0.3371%

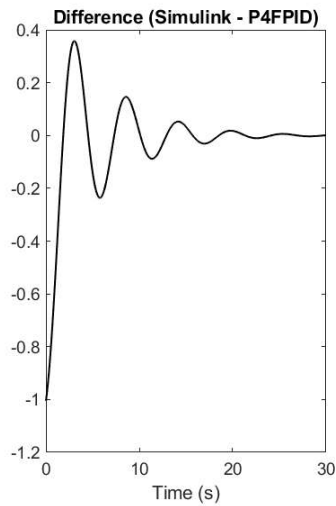
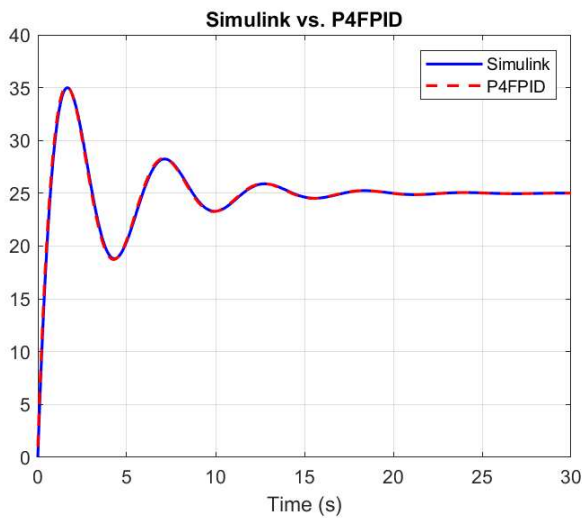
Correlation = 0.9990

Transfer Function

$$G(s) = \frac{1s + 1}{1s^2 + 1s + 1}$$

Put in: Closed-Loop

CSV File: [1.0;1.0];[1.0;1.0;1.0].csv



Results:

Fixed Step Time = 0.02 sec

Algorithm: RK-4 th order

RMSE = 0.1867

MRE = 0.5527%

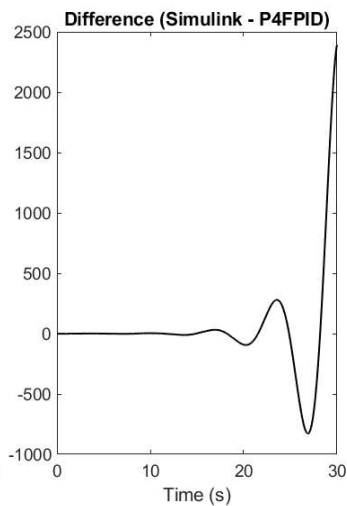
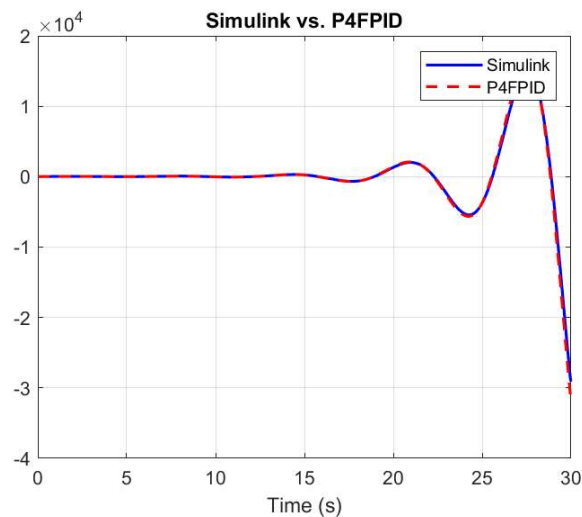
Correlation = 0.9985

Transfer Function

$$G(s) = \frac{1s^2 + 1s + 1}{1s^3 + 1s^2 + 1s + 1}$$

Put in: Closed-Loop

CSV File: [1.0;1.0;1.0];[1.0;1.0;1.0;1.0].csv



Results:

Fixed Step Time = 0.02 sec

Algorithm: RK-4 th order

RMSE = 419.9168

MRE = 15.7450%

Correlation = 0.9989

Transfer Function

$$G(s) = \frac{1s^3 + 1s^2 + 1s + 1}{1s^4 + 1s^3 + 1s^2 + 1s + 1}$$

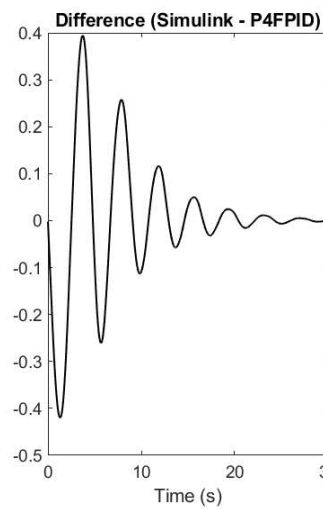
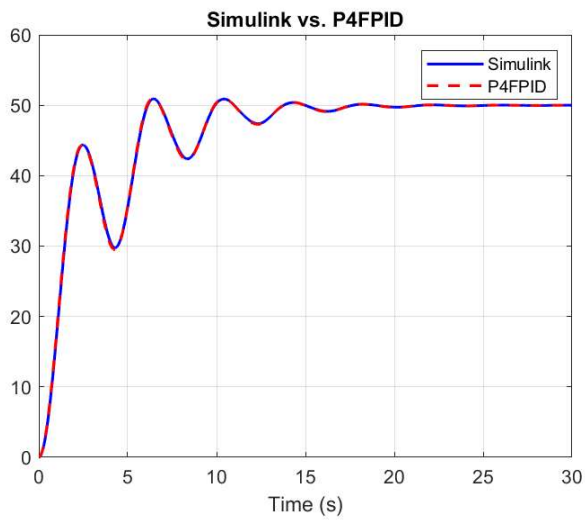
Put in: Closed-Loop

CSV File: [1.0;1.0;1.0;1.0];[1.0;1.0;1.0;1.0;1.0].csv

Discussion:

The comparative simulations reveal that the outputs generated by P4FPID and MATLAB Simulink are almost identical across all tested systems, particularly for cases involving low-scale output magnitudes. This strong agreement confirms the correctness of the implemented control algorithms and numerical procedures in P4FPID. Minor deviations observed between the two software environments are believed to originate primarily from the difference in numerical precision: P4FPID employs single-precision (float) variables to store the output vectors, whereas MATLAB relies on double-precision data types. Consequently, it is reasonable to observe that small numerical discrepancies gradually accumulate over time, especially in non-stable or highly dynamic systems, where output magnitudes amplify the accumulated error. The analysis also shows that the magnitude of these differences is inversely related to the simulation step size for instance, when a finer step such as 0.01 is used, the resulting deviations become significantly smaller. Moreover, the calculated correlation coefficients remain consistently close to unity, indicating a nearly perfect alignment between both simulation outputs. The overall pattern and dynamic behavior of the system responses are thus preserved, demonstrating that P4FPID reproduces the same temporal evolution as MATLAB Simulink. The software will be made publicly available following the publication of the associated research paper to allow further investigations, refinements of numerical methods, and extended developments by the academic and engineering community.

Fuzzy PID Controller Simulation



Results:

Fixed Step Time = 0.01 sec

Algorithm: RK-4 th order

RMSE = 0.1376

MRE = 0.6080%

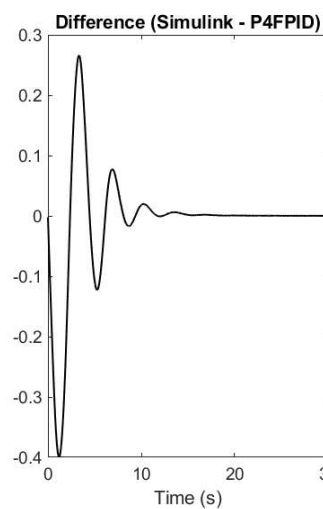
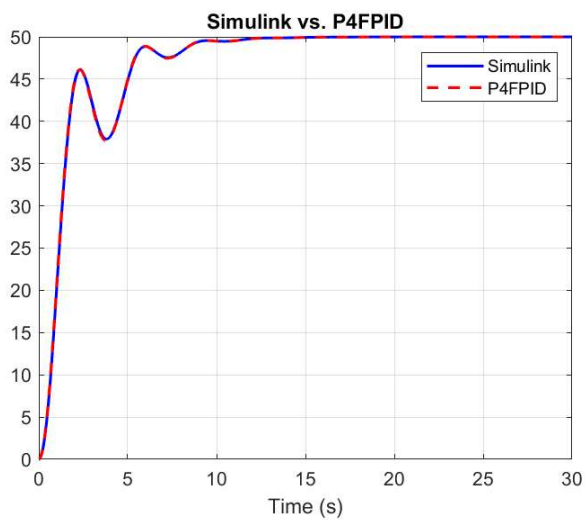
Correlation = 0.9999

Transfer Function

$$G(s) = \frac{1s + 1}{1s^3 + 1s^2 + 1s + 1}$$

Closed-Loop Fuzzy Control: fis 3*3|3

CSV File: [1.0;1.0];[1.0;1.0;1.0;1.0].csv



Results:

Fixed Step Time = 0.01 sec

Algorithm: RK-4 th order

RMSE = 0.0955

MRE = 0.4829%

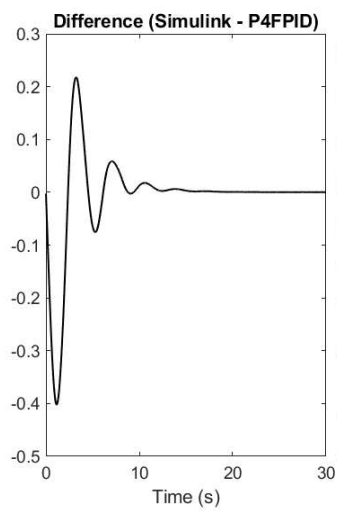
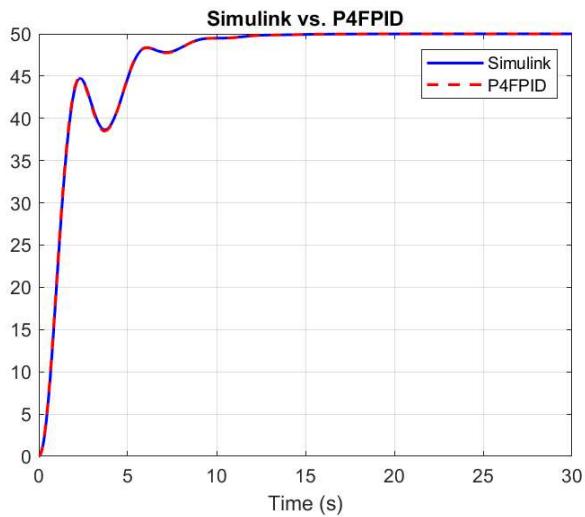
Correlation = 0.9999

Transfer Function

$$G(s) = \frac{1s + 1}{1s^3 + 1s^2 + 1s + 1}$$

Closed-Loop Fuzzy Control: fis 5*5|5

CSV File: [1.0;1.0];[1.0;1.0;1.0;1.0].csv



Results:

Fixed Step Time = 0.01 sec

Algorithm: RK-4 th order

RMSE = 0.0900

MRE = 0.4813%

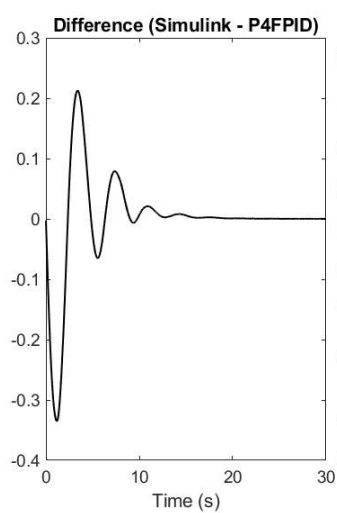
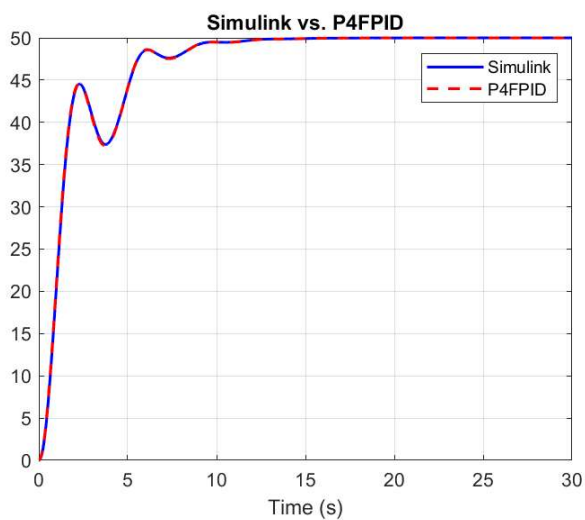
Correlation = 1.0000

Transfer Function

$$G(s) = \frac{1s + 1}{1s^3 + 1s^2 + 1s + 1}$$

Closed-Loop Fuzzy Control: fis 7*7|7

CSV File: [1.0;1.0];[1.0;1.0;1.0;1.0].csv



Results:

Fixed Step Time = 0.01 sec

Algorithm: RK-4 th order

RMSE = 0.0816

MRE = 0.4618%

Correlation = 1.0000

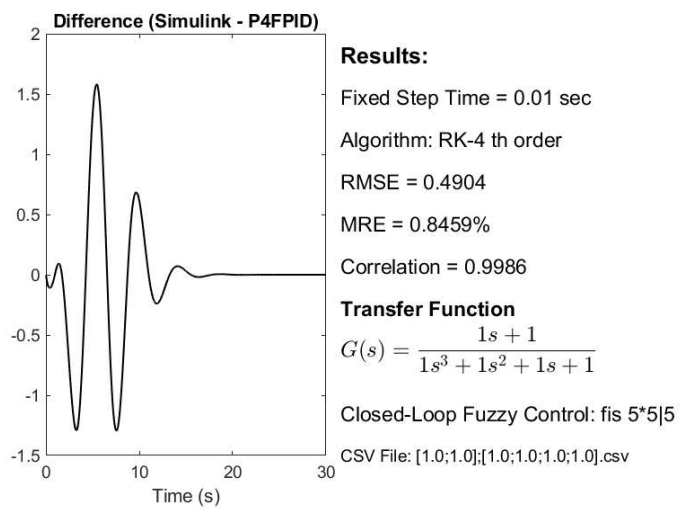
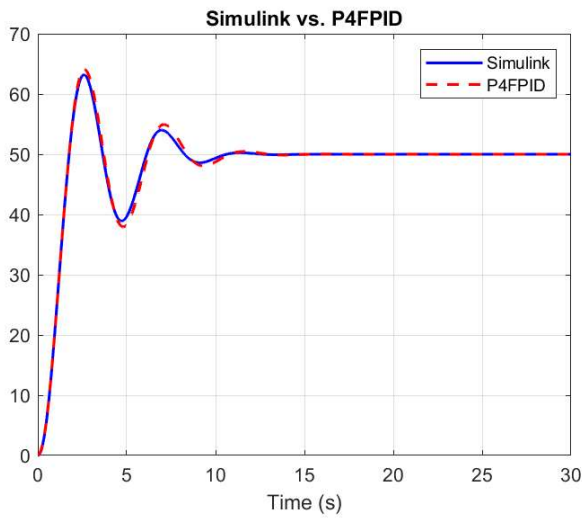
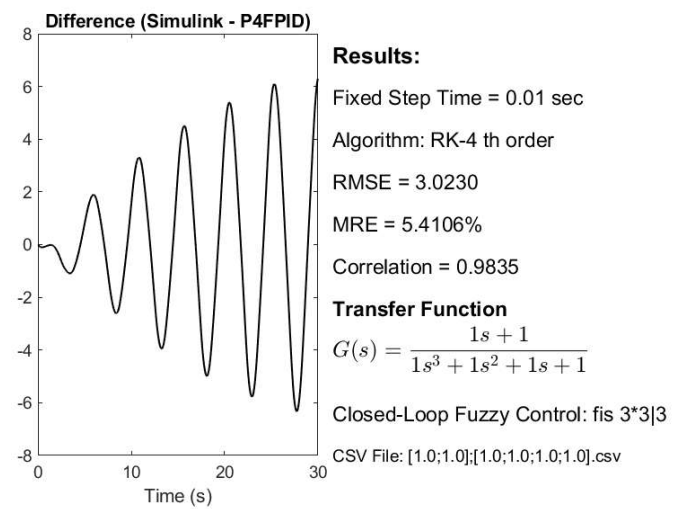
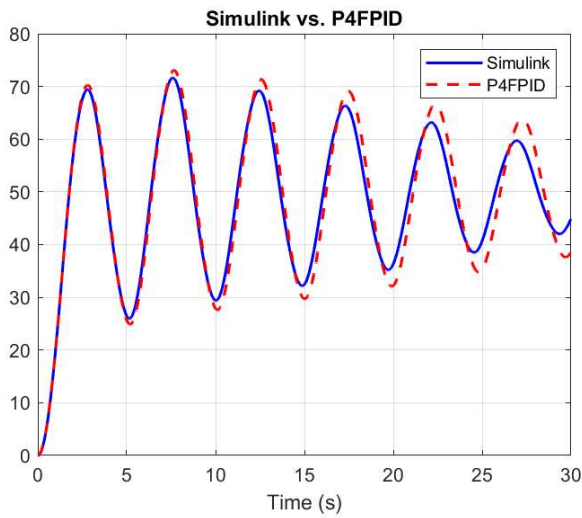
Transfer Function

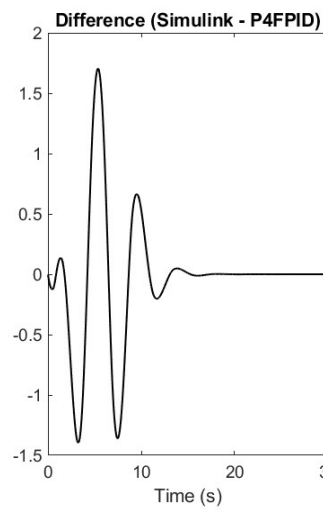
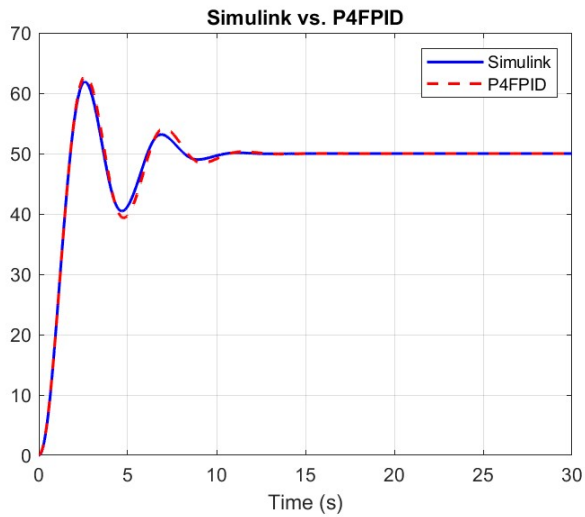
$$G(s) = \frac{1s + 1}{1s^3 + 1s^2 + 1s + 1}$$

Closed-Loop Fuzzy Control: fis 9*9|9

CSV File: [1.0;1.0];[1.0;1.0;1.0;1.0].csv

Fuzzy PD+I Controller Simulation





Results:

Fixed Step Time = 0.01 sec

Algorithm: RK-4 th order

RMSE = 0.5207

MRE = 0.8675%

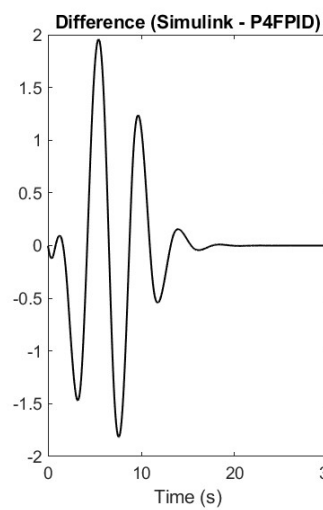
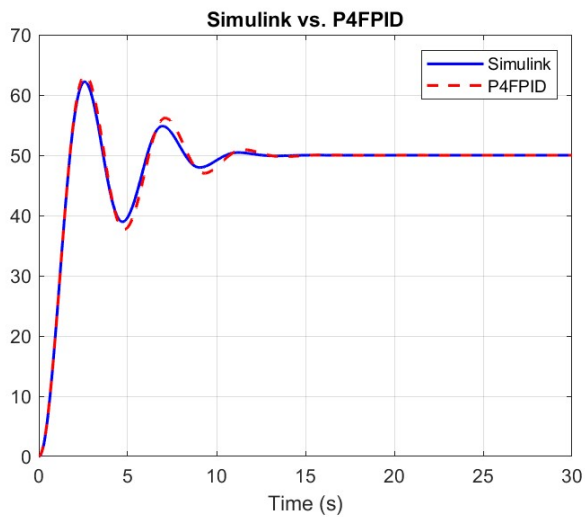
Correlation = 0.9983

Transfer Function

$$G(s) = \frac{1s + 1}{1s^3 + 1s^2 + 1s + 1}$$

Closed-Loop Fuzzy Control: fis 7*7|7

CSV File: [1.0;1.0];[1.0;1.0;1.0;1.0].csv



Results:

Fixed Step Time = 0.01 sec

Algorithm: RK-4 th order

RMSE = 0.6523

MRE = 1.0467%

Correlation = 0.9974

Transfer Function

$$G(s) = \frac{1s + 1}{1s^3 + 1s^2 + 1s + 1}$$

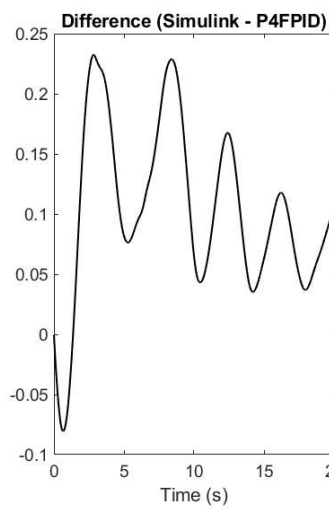
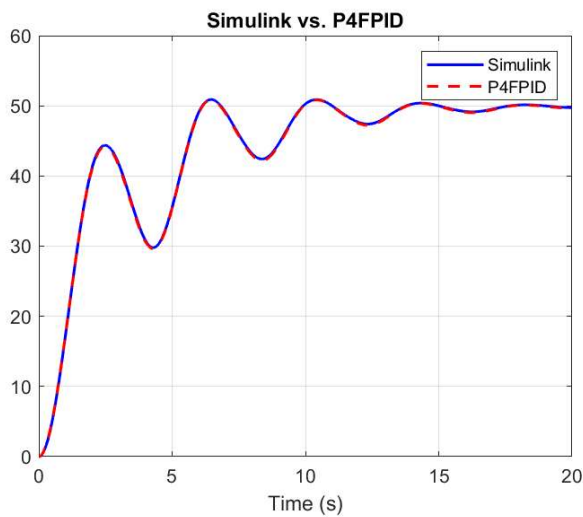
Closed-Loop Fuzzy Control: fis 9*9|9

CSV File: [1.0;1.0];[1.0;1.0;1.0;1.0].csv

Discussion

When employing fuzzy PID and fuzzy PD+I controllers, the comparative results also demonstrate a strong consistency between the outputs of P4FPID and MATLAB Simulink, particularly in systems exhibiting low oscillatory dynamics or near-stable operating conditions. The corresponding output graphs show almost identical shapes and trends, indicating that both platforms interpret the fuzzy inference mechanisms in a highly similar manner. Nevertheless, it is important to note that the controllers' internal output signals are not entirely identical. This divergence can be attributed to differences in the implementation and coding of fuzzy inference and defuzzification precision and rounding mechanisms used by each platform. Both fuzzy controllers were configured to use the same number of discretization points (which is 1000 point as indicated in the user guide manual, and also it is the standard points for Simulink fuzzy controller) for the defuzzification process, ensuring that any observed differences arise from implementation details rather than resolution disparities. Such variations, while minor in magnitude, may influence the instantaneous control actions, especially in systems with higher sensitivity or when rapid transitions occur in the input space. Despite these small discrepancies, the global system response remains coherent and dynamically equivalent across both environments, confirming the reliability of P4FPID in reproducing the behavior of established fuzzy control structures. These findings suggest that the developed tool provides a valid and transparent foundation for further exploration of fuzzy control design and hybrid tuning strategies within an open-source framework.

Benchmark Experiment



Results:

Fixed Step Time = 0.005 sec

Algorithm: RK-4 th order

RMSE = 0.1259

MRE = 0.5539%

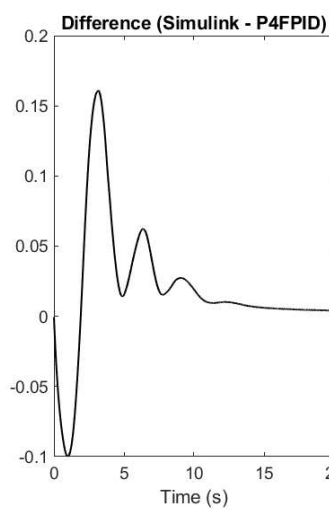
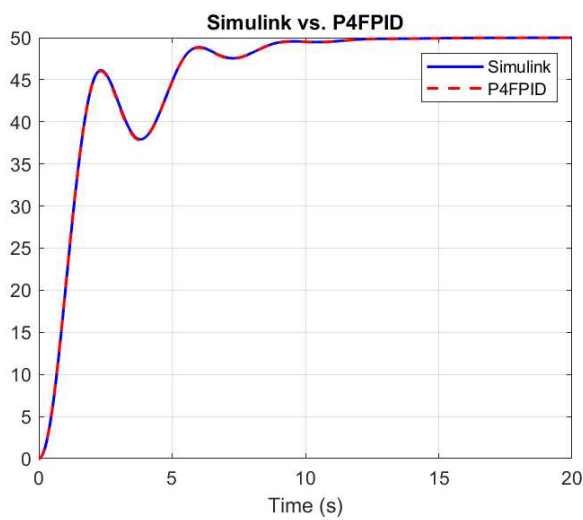
Correlation = 1.0000

Transfer Function

$$G(s) = \frac{1s + 1}{1s^3 + 1s^2 + 1s + 1}$$

Closed-Loop Fuzzy PID: fis 3*3|3

CSV File: [1.0;1.0];[1.0;1.0;1.0;1.0].csv



Results:

Fixed Step Time = 0.005 sec

Algorithm: RK-4 th order

RMSE = 0.0503

MRE = 0.3658%

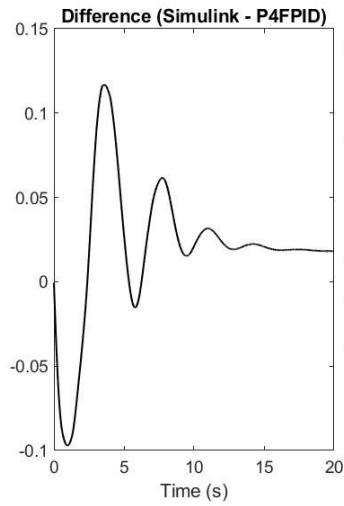
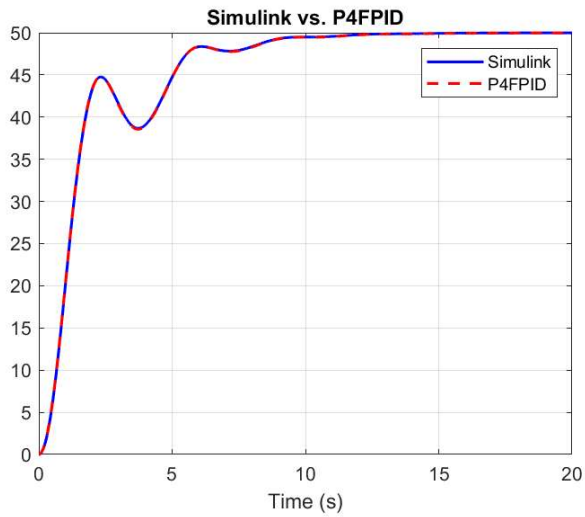
Correlation = 1.0000

Transfer Function

$$G(s) = \frac{1s + 1}{1s^3 + 1s^2 + 1s + 1}$$

Closed-Loop Fuzzy PID: fis 5*5|5

CSV File: [1.0;1.0];[1.0;1.0;1.0;1.0].csv



Results:

Fixed Step Time = 0.005 sec

Algorithm: RK-4 th order

RMSE = 0.0462

MRE = 0.3809%

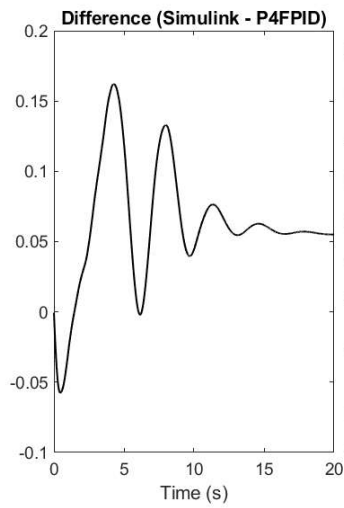
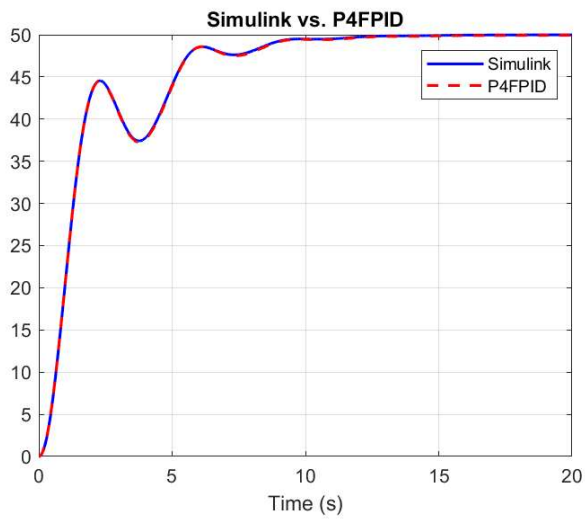
Correlation = 1.0000

Transfer Function

$$G(s) = \frac{1s + 1}{1s^3 + 1s^2 + 1s + 1}$$

Closed-Loop Fuzzy PID: fis 7*7|7

CSV File: [1.0;1.0];[1.0;1.0;1.0;1.0].csv



Results:

Fixed Step Time = 0.005 sec

Algorithm: RK-4 th order

RMSE = 0.0751

MRE = 0.4196%

Correlation = 1.0000

Transfer Function

$$G(s) = \frac{1s + 1}{1s^3 + 1s^2 + 1s + 1}$$

Closed-Loop Fuzzy PID: fis 9*9|9

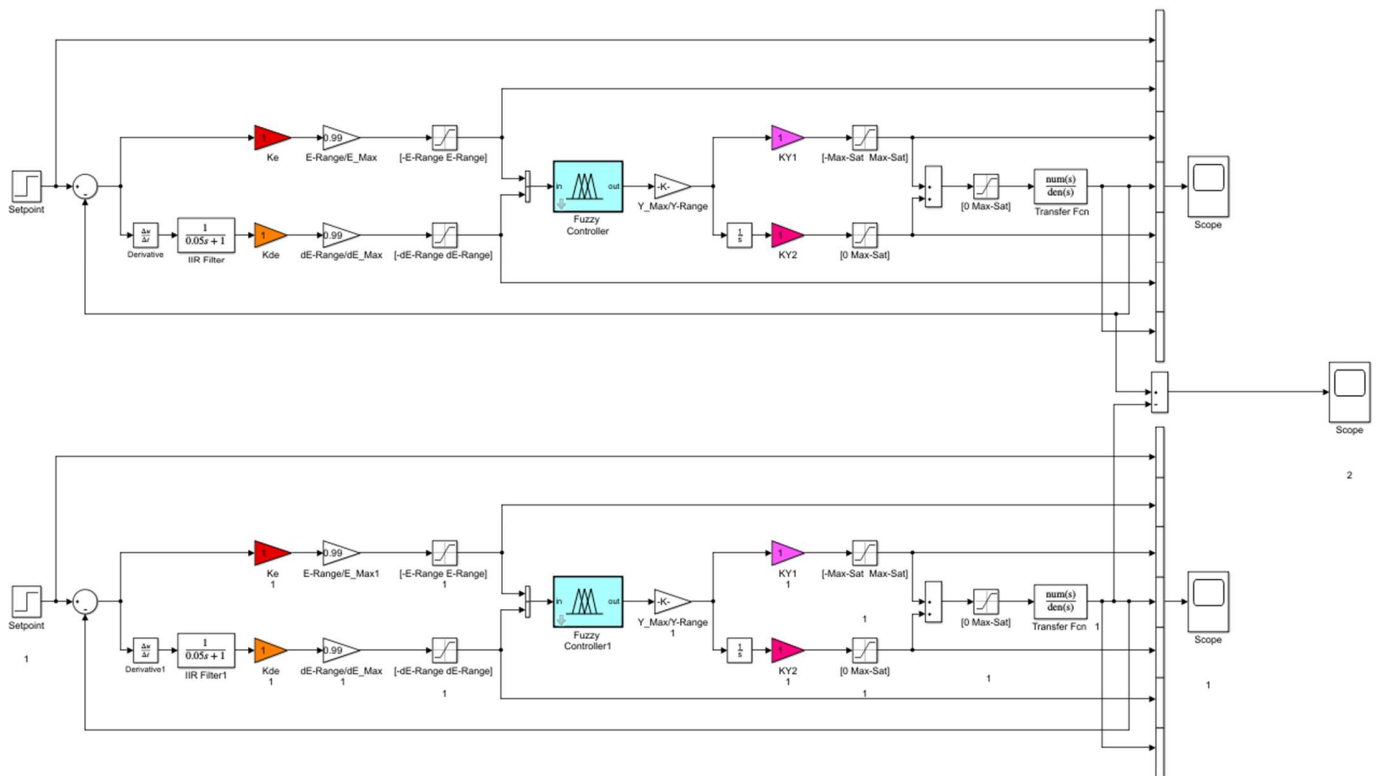
CSV File: [1.0;1.0];[1.0;1.0;1.0;1.0].csv

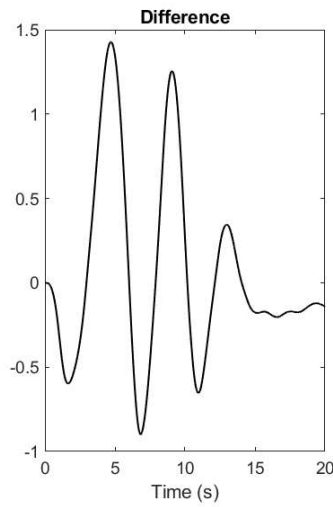
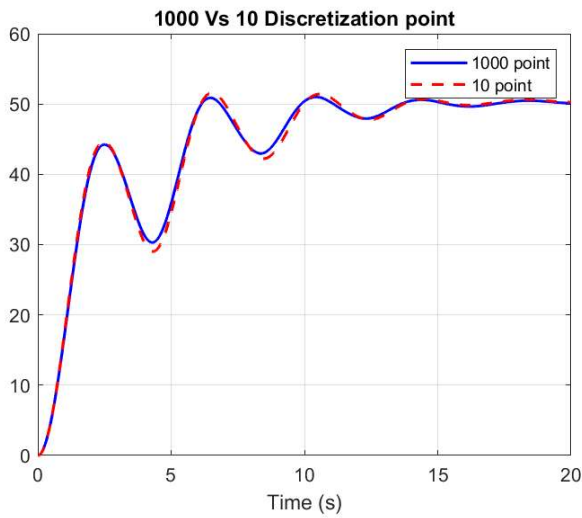
Discussion

In the benchmarking experiment, the comparison between P4FPID and MATLAB Simulink confirms an almost perfect agreement between the resulting output graphs. The figures illustrate that the two systems produce entirely identical responses, with only negligible errors that can be considered practically insignificant. To ensure the highest level of numerical precision and reproducibility, the number of discretization points was set to the default values which is 1000 point, the simulation step size was fixed at 0.005, a value carefully selected to minimize numerical drift and rounding effects. This fine discretization allowed both software environments to operate under equivalent computational conditions, thereby validating the accuracy and consistency of the implemented algorithms. The nearly indistinguishable plots provide strong evidence that P4FPID is capable of replicating the exact system dynamics as obtained in MATLAB Simulink, confirming that the developed platform offers reliable numerical fidelity and trustworthy simulation outcomes suitable for control-system research and instructional use.

Further investigations:

A complementary simulation investigation was conducted to examine the influence of varying the number of discretization points within MATLAB Simulink's fuzzy logic engine. Specifically, the system outputs were compared under two configurations: the default discretization of 1,000 points and an alternative configuration using only 10 points, as illustrated in the Figure below. The choice of 10 discretization points was made because it represents the lowest number that still produced acceptable differences in the system's output compared to the default configuration.





Results:

Fixed Step Time = 0.005 sec

Algorithm: RK-4 th order

RMSE = 0.5649

MRE = 1.0604%

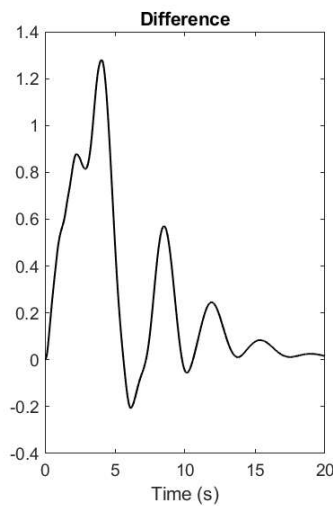
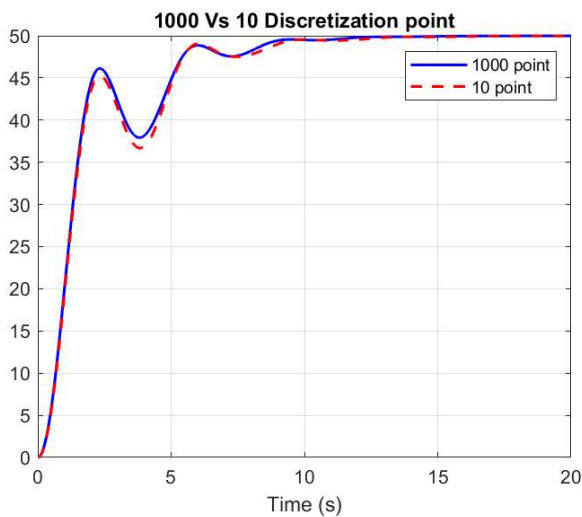
Correlation = 0.9987

Transfer Function

$$G(s) = \frac{1s + 1}{1s^3 + 1s^2 + 1s + 1}$$

Closed-Loop Fuzzy PID: fis 3*3|3

CSV File: [1.0;1.0];[1.0;1.0;1.0;1.0].csv



Results:

Fixed Step Time = 0.005 sec

Algorithm: RK-4 th order

RMSE = 0.4434

MRE = 0.8378%

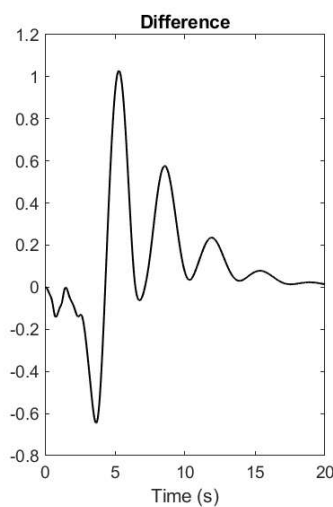
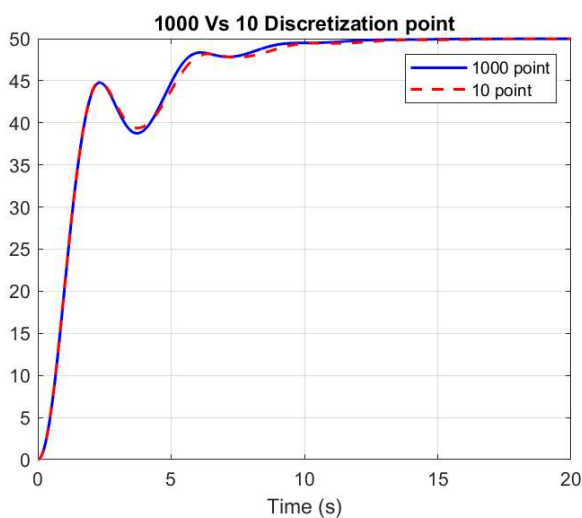
Correlation = 0.9994

Transfer Function

$$G(s) = \frac{1s + 1}{1s^3 + 1s^2 + 1s + 1}$$

Closed-Loop Fuzzy PID: fis 5*5|5

CSV File: [1.0;1.0];[1.0;1.0;1.0;1.0].csv



Results:

Fixed Step Time = 0.005 sec

Algorithm: RK-4 th order

RMSE = 0.3081

MRE = 0.4682%

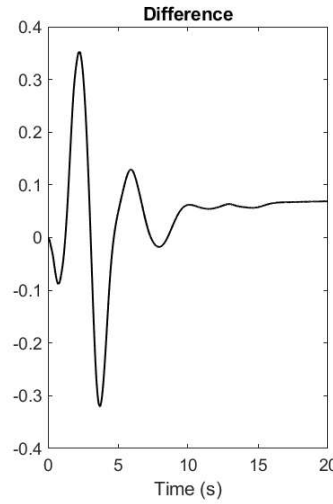
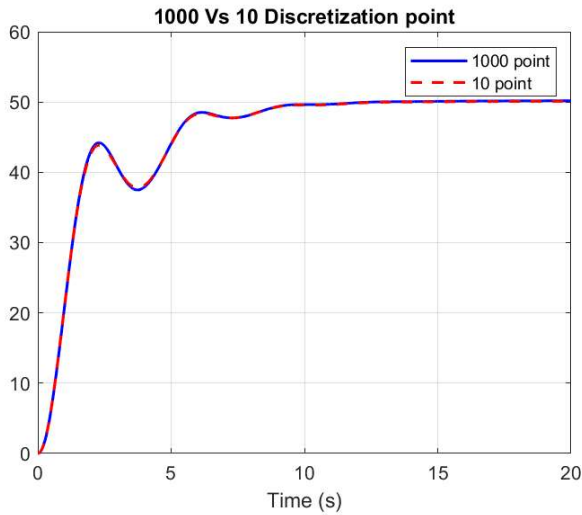
Correlation = 0.9996

Transfer Function

$$G(s) = \frac{1s + 1}{1s^3 + 1s^2 + 1s + 1}$$

Closed-Loop Fuzzy PID: fis 7*7|7

CSV File: [1.0;1.0];[1.0;1.0;1.0;1.0].csv



Results:

Fixed Step Time = 0.005 sec

Algorithm: RK-4 th order

RMSE = 0.1117

MRE = 0.2277%

Correlation = 1.0000

Transfer Function

$$G(s) = \frac{1s + 1}{1s^3 + 1s^2 + 1s + 1}$$

Closed-Loop Fuzzy PID: fis 9*9|9

CSV File: [1.0;1.0];[1.0;1.0;1.0;1.0].csv

The obtained results revealed that MATLAB Simulink's internal computations exhibited only minor variations between the two settings. All previously mentioned .fis configurations were employed to ensure consistent comparison conditions. From an experimental standpoint, the observed differences remain within an acceptable range. This additional analysis was performed to establish a fair benchmarking framework between our software, P4FPID (tested under its highest computational load), and MATLAB Simulink (tested under its lowest computational load). The motivation behind this step lies in ensuring objectivity and transparency in our evaluation, especially considering that MATLAB's internal fuzzy logic implementation operates as a closed-source (black-box) system, limiting the ability to fully interpret its internal architecture.